BASIC 9

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STRAND 1: INTRODUCTION TO COMPUTING

SUB-STRAND 1: COMPONENTS OF COMPUTERS AND COMPUTER SYSTEMS

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.1.1.1 Identify parts of a Computer and Technology Tools	**B91.1.1.1. Discuss the trends in the next generation of computers** **Exemplar(s):** 1. Identify features expected to be seen in the next generation of computers. 2. Describe Google quantum computer (sycamore) and compare its processing power with other supercomputers (focus on processor).	**Communication and Collaboration (CC)** **CC7.5**: Identify and analyse differentpoints of views of speaker. **CC7.3**: Provide feedback in areas of ideas, organisation, voice, word choice and sentence fluency in communication.
	B9.1.1.1.2. Examine the concept of Perceptual Computing **Exemplar(s):** 1. Discuss the features of Perceptual Computing.	**CC7.5**: Identify and analyse differentpoints of views of speaker. **CC7.3**: Provide feedback in areas of ideas, organisation, voice, word choice and sentence fluency in communication.
	B9.1.1.1.3 Discuss the uses of Output devices such as WearableDisplays, E-Paper, E-Books, Kindle **Exemplar(s):** 1. Describe Wearable Displays (e.g. Google Glass), E-Paper, E-Books (e.g.Kindle)	**Communication and Collaboration (CC), Digital Literacy (DL)** **CC7.5**: Identify and analyse differentpoints of views of speaker.
	B9.1.1.1.4 Describe Storage Systems: Network and Cloud Storage Systems, Smart Cards, Holographic Storage, Storage Systems for Large Computer Systems (home servers or media servers) **Exemplar(s):** 1. Discuss Network and Cloud Storage Systems, Smart Cards, Holographic Storage, Storage Systems for Large Computer Systems (home servers or media servers). 2. Explore common cloud storage examples such as Google Drive, One Drive, etc. 3. Discuss the pros and cons of using cloud storage.	**DL5.3:** Ability to find and utilise digital content. **CC7.5**: Identify and analyse differentpoints of views of speaker.

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CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.1.1.2. Demonstrate the Use of the Desktop	**B91.1.2.1 Explore personalisation of the computer** **Exemplar(s):** 1. Change desktop icons. 2. Change mouse pointers.	**Communication and Collaboration (CC), Digital Literacy (DL)** **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem.
	B9.1.1.2.2 Identify and use file management techniques (drivers and hardware) **Exemplar(s):** 1. Identify and explore the use of device drivers e. g. sound drivers and video graphic drivers 2. Explain plug-and-play devices. 3. Demonstrate how to install, update or delete drivers.	**CC7.5**: Identify and analyse differentpoints of views of speaker. **DL5.3:** Ability to find and utilise digital content.

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STRAND 1: INTRODUCTION TO COMPUTING

SUB-STRAND 2: TECHNOLOGY INTHE COMMUNITY (COMMUNICATION)

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.1.2.1. Demonstrate the Use of Technology in the Community	**B9.1.2.1.1. Evaluate problems in the community that can be solvedwith technology** **Exemplar(s):** 1. Discuss problems in the community that can be solved using technology.Work in pairs.	**Creativity and Innovation (CI), Communication and Collaboration (CC), Critical Thinking and ProblemSolving (CP).** **CC8.1:** Speak clearly and explain ideas. Share a narrative or extended answer while speaking to a group. **CC9.2**: Understand and useinterpersonal skills. **CP5.1:** Ability to combine information and ideas from several sources to reach a conclusion.
	B9.1.2.1.2. Propose solutions to the problems identified **Exemplar(s):** 1. Present technological solutions to the problems identified in B9.1.2.1.1.	**CC8.1:** Speak clearly and explain ideas. Share a narrative or extended answer while speaking to a group. **CP5.1:** Ability to combine information and ideas from several sources to reach a conclusion.
	B9.1.2.1.3. Design the solution selected **Exemplar(s):** 1. Design the solution selected. 2. Present the solution designed. ***NB: Tasks in B9.1.2.1 is to be done as a project work***	**CI 5.7:** Putting forward constructive comments, ideas, explanations and newways of doing things. **CC8.1:** Speak clearly and explain ideas.Share a narrative or extended answer while speaking to a group.

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STRAND 1: INTRODUCTION TO COMPUTING

SUB-STRAND 3: HEALTH AND SAFETY INTHE USE OF ICT TOOLS

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.1.3.1. Demonstrate How to Apply Health and Safety Measures in the Use ICT Tools	**B9.1.3.1.1 Evaluate health issues at workstationsExemplar(s):** 1. Discuss the use of ergonomic tools such as ergonomic keyboard and paperstand. 2. Demonstrate the appropriateness of the lighting system while working withthe computer.	**CC8.1:** Speak clearly and explain ideas. Share a narrative or extended answer while speaking to a group.
	B9.1.3.1.2 Evaluate Safety Risk Reduction at issues at workstations Exemplar(s): 1. Discuss the heat generated by machines (e.g. computers, printers etc.) andhow to reduce excess heat.	**CC8.1:** Speak clearly and explain ideas. Share a narrative or extended answer while speaking to a group.

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STRAND 2: PRODUCTIVITY SOFTWARE

SUB-STRAND 1: INTRODUCTION TO WORD PROCESSING

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.2.1.1 Demonstrate How to Use Microsoft Word (Multimedia)	**B9.2.1.1.1. Demonstrate how to add pictures, insert a screenshot and screen clipping and print screen** **Exemplar(s):** 1. Explore the use of the clip art, screenshot and screen clipping in the Insert Ribbon. 2. Demonstrate the use of the print screen key in capturing and inserting pictures.	**Creativity and Innovation (CI), Digital Literacy (DL)** **CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges. **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem.
	B9.2.1.1.2. Demonstrate the use of SmartArt **Exemplar(s):** 1. Illustrate the use of SmartArt in the Illustrations group of the Insert Ribbon.	**CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges. **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem.
	B9.2.1.1.3. Demonstrate how to add Multimedia (audios, videos, animations), Charts and Hyperlinks **Exemplar(s):** 1. Explore the use of the clip art and screenshot in the Insert Ribbon.	**CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges. **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem.

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STRAND 2: PRODUCTIVITY SOFTWARE

SUB-STRAND 2: INTRODUCTION TO PRESENTATION

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.2.2.1 DemonstrateHow to use Microsoft PowerPoint (Multimedia)	**B9.2.2.1.1. Demonstrate how to add pictures and insert screenshots** **Exemplar(s):** 1. Explore the use of the clip art and screenshot in the Insert Ribbon.	**Creativity and Innovation (CI),Digital Literacy (DL)** **CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem
	B9.2.2.1.2. Demonstrate how to animate slides in a presentation **Exemplar(s):** 1. Demonstrate the use of transitions and animations. 2. Create a seven-slide presentation with animations and transitions.	**CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem
	B9.2.2.1.3. Demonstrate how to add Multimedia (audios, videos etc.), tables and charts **Exemplar(s):** 1. Explore the use of the Insert Ribbon tab to add multimedia (e.g. audios, videos, etc.). 2. Demonstrate the use of tables and charts in slides.	**CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem

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STRAND 2: PRODUCTIVITY SOFTWARE

SUB-STRAND 3: INTRODUCTION TO DESKTOP PUBLISHING

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.2.3.1. Critique a Desktop Published Document	**B9.2.3.1.1 Create and present a desktop published document (flyer, advertisement, invitation cards, business cards)** **Exemplar(s):** 1. Create and present a four-page document with images and overflow e.g.flyer, advertisement, storybook.	**Creativity and Innovation (CI), Communication and Collaboration (CC), Digital Literacy (DL), Critical Thinking and Problem Solving (CP).** **CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem. **CC9.6:** Ability to work with all group members to complete a task successfully.
	B9.2.3.1.2 Describe a desktop published document **Exemplar(s):** 1. Give out your document to another learner assigned to you and collect the learner's document. 2. Examine the document received in terms of the position of text and images,use of colour, mechanics, content accuracy, etc.	**DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem. **CP5.1:** Ability to combine information and ideas from several sources to reach a conclusion **CC9.6:** Ability to work with all group members to complete a task successfully

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CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.2.3.1. Critique a Desktop Published Document	**B9.2.3.1.3 Evaluate a desktop published documentExemplar(s):** 1. Evaluate the assigned document using, but not limited to the following criteria: position of text and images, general layout of the document, use of colour, mechanics (punctuation, spelling, italics, capitalisation, etc.), andappropriateness of the design for the intended purpose.	**CC9.6:** Ability to work with all group members to complete a task successfully. **CP5.1:** Ability to combine information and ideas from several sources to reach a conclusion.

STRAND 2: PRODUCTIVITY SOFTWARE

SUB-STRAND 4: INTRODUCTION TO ELECTRONIC SPREADSHEET

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.2.4.1. Demonstrate How to Use Spreadsheet (Advanced Operations)	**B9.2.4.1.1 Perform data filtering, sorting and validation** **Exemplar(s):** 1. Describe Fields (columns), Records (rows) and Tables (structured/ unstructured). 2. Illustrate examples of structured and unstructured tables. 3. Construct a structured data table of class members (e.g. data table may havethe following fields: Surname, First name, Date of Birth, Sex, Home Town, Region, etc). 4. Apply validation rules to check for errors. 5. Convert data tables to a list in MS Excel. 6. Demonstrate the process of entering data to a list. 7. Demonstrate the use of validation list in the Sex column (in Exemplar(s): 3)to provide options in a drop-down for data entry. 8. Demonstrate how to sort data in alphabetical order (ascending/descending)and filter data to display only selected data.	**Communication and Collaboration (CC), Digital Literacy (DL), Critical Thinking and Problem Solving (CP).** **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem **CP5.1:** Ability to combine information and ideas from several sources to reach a conclusion **CC9.6:** Ability to work with all group members to complete a task successfully

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CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.2.4.1. Demonstrate how to use spreadsheet (Advanced Operations)	**B9.2.4.1.2. Demonstrate how to use styles, themes, templates and macros** **Exemplar(s):** 1. Describe the difference between templates and macros. 2. Create new spreadsheet documents from predefined templates in MS Excel. 3. Demonstrate the use of styles and themes on sample worksheets. 4. Do a project on formatting a dataset by applying styles and themes. 5. Explore the use of macros.	**Creativity and Innovation (CI), Communication and Collaboration (CC), Digital Literacy (DL), Critical Thinking and Problem Solving (CP).** **CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges. **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem. **CC9.6:** Ability to work with all group members to complete a task successfully.
	B9.2.4.1.3. Demonstrate the use of data tables, pivot tables, charts and pivot charts **Exemplar(s):** 1. Explore pivot tables and charts. 2. Explore the use of a pivot table to display a summary of the dataset (refer to indicator **B9.2.4.1.1**). 3. Insert a pivot chart to display the number of males and females in the class. 4. Demonstrate the use of the sort and filter features of the pivot table.	**CC9.6:** Ability to work with all group members to complete a task successfully. **CP5.1:** Ability to combine information and ideas from several sources to reach a conclusion.

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STRAND 3: COMMUNICATION NETWORKS

SUB-STRAND 1: COMPUTER NETWORKS

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.3.1.1. Know the Concept of Computer Networking for Global Communications	**B9.3.1.1.1 Discuss types of e-commerce and the cashless society (Bitcoin, Transaction cards, Quick Response code (QR) payment system)** **Exemplar(s):** 1. Explore the use of the Internet to engage in online business: selling, buyingand paying for products online. 2. Discuss the use of online banking systems (e.g. using mobile money, bitcoin,the use of MasterCard, Visa card, QR code payment system, etc.).	**Digital Literacy (DL), Communication and Collaboration (CC)** **DL5.3:** Ability to find and utilise digital content **CC8.5**:Vary the level of detail and the language used when presenting to make it appropriate to the audience
	B9.3.1.1.2 Justify eLearning potentials **Exemplar(s):** 1. Explain the concept of eLearning, its benefits and disadvantages. 2. Discuss projects on a collaborative platform (e.g. iBox network, the use ofad hoc network to share resources, Wikis, Google Docs] etc.)	**CC8.1:** Speak clearly and explain ideas. Share a narrative or extended answer while speaking to a group **DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem

STRAND 3: COMMUNICATION NETWORKS

SUB-STRAND 2: INTERNET AND SOCIAL MEDIA

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.3.2.1 Demonstrate the Use of Social Networking and Electronic Mail	**B9.3.2.1.1 Identify the advantages and risks in the use of social media platforms** **Exemplar(s):** 1. Illustrate the benefits of using social media sites 2. Discuss the issues and risks surrounding the usage of social media platformsand how to avoid them. 3. Explore reply, reply all, forward and forward all features in the use of emails	**DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem.

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STRAND 3: COMMUNICATION NETWORKS

SUB-STRAND 3: INFORMATION SECURITY

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.3.3.1. Recognise data threats and the means of protection	**B9.3.3.1.1 Discuss cyberbullying, cyberstalking, digital footprint anddigital shadow on the Internet** **Exemplar(s):** 1. Watch a film or do a reading on cyberbullying, cyberstalking, digital footprint,digital shadows. 2. Discuss the nature of cyberbullying, cyberstalking, digital footprint and digital shadows. 3. Identify examples of cyberbullying, cyberstalking, digital footprint and digital shadows. 4. Discuss the effects on cyberbullying on individuals	**Creativity and Innovation (CI), Digital Literacy (DI), Critical Thinking and Problem Solving (CP), Communication and Collaboration (CC)** **DL 6.1:** Understand the sociological and emotional aspects of cyberspace **CP 6.4:** Ability to identify important and appropriate criteria and use them to evaluate available alternatives **CC8.5**: Vary the level of detail and the language used when presenting to make it appropriate for the audience
B9.3.3.1. Recognise data threats and the means of protection	**B9.3.3.1.2 Explain ten (10) information hacking techniques on the Internet environment.** **Exemplar(s):** 1. Brainstorm information hacking techniques on the internet environment. 2. Explain ten (10) information hacking techniques e.g. phishing, keyloggers,Denial of Service attack, eavesdropping, etc.	**DL5.1:** Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectivelyuse it to solve a problem .

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STRAND 3: COMMUNICATION NETWORKS

SUB-STRAND 4: WEB TECHNOLOGIES

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.3.4.1 Demonstratethe Use of a Web Browser (Blogging)	**B9.3.4.1.1 Examine the importance of creating blogs** **Exemplar(s):** 1. Discuss the importance of creating blogs.	**Creativity and Innovation (CI),Digital Literacy (DL).** **CI 6.6:** Being open-minded, adapting and modifying ideas to achieve creativeresults. **DL5.3:** Ability to find and utilise digital content
	B9.3.4.1.2 Develop a blog for the school or a social club **Exemplar(s):** 1. Investigate the items to include in a school or social club blog. 2. Develop a blog for the school or a social club.	**DL6.3:** Use digital tools to create novel things
	B9.3.4.1.3 Explore the steps in publishing a blogExemplar(s): 1. Identify steps in publishing a blog. 2. Demonstrate the procedure for publishing a blog and invite others to comment.	**DL6.3:** Use digital tools to create novel things

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STRAND 4: COMPUTATIONAL THINKING

SUB-STRAND 1: INTRODUCTION TO PROGRAMMING

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.4.1.1. Show an Understanding of the Concept of Programming	**B9.4.1.1.1 Describe the conversion of decimal into binary data typefor computer to recognise the meaning, process and store** **Exemplar(s):** 1. Convert decimal, binary and hexadecimal data from one format to another. 2. Show the results of calculating two or more binary numbers using the mathematical notation or operators in the number base two rule.	**Creativity and Innovation (CI),Critical Thinking and ProblemSolving (CP).** **CP 5.5:** Effectively evaluate the success of solutions used in an attempt to solve a complex problem
	B9.4.1.1.2 Identify the different tools which are accessible in Integrated Development Environment (IDE) to aid the development of codes **Exemplar(s):** 1. Explore programming languages such as Snap, Scratch and Python to explainthe key terminologies (variables, operators, controls, events, etc.) around the coding environment. Explore a web development programme to create a simple website.	**CI 5.5:** Ability to try new alternatives and different approaches. **CP 6.6:** Preparedness to recognise andexplain results after implementation ofplans.

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STRAND 4: COMPUTATIONAL THINKING

SUB-STRAND 2:ALGORITHM

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.4.2.1.Analyse the Correct Step-by-step Procedure in Solving any Real-world Problem	**B9.4.2.1.1 Write a programme using flowchart and Pseudocode algorithm that includes sequence, selection and iteration choices in problem-solving** **Exemplar(s):** 1. Discuss at least three ways to do proper hand-washing and prepare beverages with or without sugar and/or milk. 2. Write an algorithm for Exemplar(s): 1 that focuses on procedure correctness and shortest time to execute.	**Creativity and Innovation (CI),Critical Thinking and ProblemSolving (CP).** CI6.1: Exhibit strong memory, intuitive thinking; and respond appropriately. **CP 6.4:** Ability to identify important and appropriate criteria and use them to evaluate available alternatives.
	B9.4.2.1.2 Translate a Flowchart algorithm to Pseudocode format andvice versa **Exemplar(s):** 1. Write an algorithm using flowchart format. Convert or translate the same flowchart algorithm into a Pseudocode format. (do a vice versa translation example to reinforce critical thinking)	**CI5.4:** Ability to visualise alternatives, see possibilities and identify problems and challenges.

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STRAND 4: COMPUTATIONAL THINKING

SUB-STRAND 3: ROBOTICS

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.4.3.1. Discuss Robot Intelligence Concepts	**B9.4.3.1.1 Construct a robot artefact using available lab componentsand tools or emulator/simulator software pack.** **Exemplar(s):** 1. Explain the three basic laws of robotics by Isaac Asimov i.e. Asimov's science fiction laws of robotics. 2. Demonstrate how a robot is assembled using real robots' toolkit/video/ pictures. 3. Explore a robotic software pack, e.g. Scratch, Webot, Snap, Mbot software, EV3 and Mobile Applications such as lightbot	**Creativity and Innovation (CI),Critical Thinking and ProblemSolving (CP).** **CI 6.3:** Ability to select the most effective creative tools for work and givereasons for the choice **CP 5.4:** Generate hypothesis to help answer complex problems

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STRAND 4: COMPUTATIONAL THINKING

SUB-STRAND 4:ARTIFICIAL INTELLIGENCE

CONTENT STANDARD	INDICATORS AND EXEMPLAR(S)	CORE COMPETENCIES
B9.4.4.1 Discuss Artificial intelligence Concepts	**B9.4.4.1.1. Describe the knowledge-based systems (Expert systems)as the classical Artificial intelligence** **Exemplar(s):** 1. Illustrate the use of IF-THEN control structure for querying an expert system 2. Demonstrate how to input a request in any knowledge-based system to generate an output or result (e.g. Telemedicine system) 3. Demonstrate how to go onto the web and use Google's Teachable Machine demo to get a basic understanding of how machine learning works (e.g. Whatisit as an open-source cloud-based app which can identify the object inan image/photo) NB: Demonstrate how data is collected and the extent to which information can be used. Also discuss thoughts on machine learning. https:// teachablemachine.withgoogle.com	**Communication and Collaboration (CC), Critical Thinking and ProblemSolving (CP)** **CC8.2**: Explain ideas in a clear order with relevant detail, using correct construction and structure of speech. **CP 6.6:** Preparedness to recognise and explain results after implementation of plans.

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GLOSSARY

1.	**Holographic storage**	It is a computer storage device that uses beams of light to store three-dimensional digitally created data.
2.	**Cloud Storage**	Includes saving data in a remote physical location that can be accessed through the Internet from any device.
3.	**Wearable computer**	It is a technology tool that a customer can wear usually to monitor health and fitness-related information.
4.	**E-Paper**	It is a technology that makes this easy to read text on an electronic gadget.
5.	**E-Books**	An electronic version of a physical book that can be read on a screen or a handheld device specifically designed forreading.
6.	**Kindle**	This is the first portable hardware computer released in the U.S. Kindle shows images and text using E ink, and can useSprint's Evolution-Data Optimised (EVDO) to get e-books over the Amazon Whispernet.
7.	**Smart Cards**	A plastic card with a built-in microprocessor, used typically to perform financial transactions.
8.	**Sensors**	A device which detects or measures a physical property and records, indicates, or otherwise responds to it.
9.	**BODMAS**	This is an acronym and it stands for Bracket, Of, Division, Multiplication, Addition and Subtraction. It is an order of operations which includes a collection of rules that reflect conventions about which procedures to perform first inorder to evaluate a given mathematical expression.
10.	**Quantum**	Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. A quantum computer is used to perform such computation, which can be implemented theoretically or physically.
11.	**Sycamore**	Sycamore is the name of Google's quantum processor comprising 54 qubits (quantum bits). Sycamore achieved itsresults using exactly 53 qubits. A 54th one on the chip failed.

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APPENDICES

APPENDIX I: CORE COMPETENCIES AND SUBSKILLS OF THE COMMON CORE PROGRAMME (CCP)

1. COMMUNICATION AN COLLABORATION (CC)

B7-B9		
CC7: LISTENING	**CC8: PRESENTING**	**CC9: TEAMWORK**
CC7.1: Identify words or sentences in context appropriately	**CC8.1**: Speak clearly and explain ideas. Share a narrative or extended answer while speaking to agroup	**CC9.1**: Demonstrate behaviour and skills ofworking towards group goals
CC7.2: Interpret correctly and respond to non-verbal communication such as facial expressions,cues and gestures	**CC8.2**: Explain ideas in a clear order with relevant detail, using correct construction and structure of speech	**CC9.2**: Understand and use interpersonal skills
CC7.3: Provide feedback in areas of ideas, organisation, voice, word choice and sentence fluency in communication	**CC8.3**: Apply appropriate diction, and structure sentences correctly for narrative, persuasive, imaginative and expository purposes	**CC9.3**: Understand roles during group activities
CC7.4: Identify underlying themes, implicationsand issues when listening	**CC8.4**: Anticipate different responses from the audience and plan for them	**CC9.4**: Help group work on relevant activities
CC7.5: Identify and analyse different points ofviews of speaker	**CC8.5**: Vary the level of detail and the languageused when presenting to make it appropriate to the audience	**CC9.5**: Appreciate the importance of includingall team members in discussions and actively encourage contributions from them
		CC9.6: Ability to work with all group members to complete a task successfully
		CC9.7: Effectively perform multiple roles within the group
		CC9.8: Demonstrate an awareness of the wider team dynamics and work to minimise conflicts inthe team

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2. CRITICAL THINKING AND PROBLEM SOLVING (CP)

B7-B9	
CP5: CRITICAL THINKING	**CP6: PROBLEM SOLVING**
CP 5.1: Ability to combine information and ideas from several sources toreach a conclusion	**CP 6.1:** Ability to effectively define goals towards solving a problem
CP 5.2: Analyse and make distinct judgement about viewpoints expressed inan argument	**CP 6.2:** Ability to explain plans for attaining goals
CP 5.3: Create simple logic trees to think through problems	**CP 6.3:** Identify important and appropriate alternatives
CP 5.4: Generate hypothesis to help answer complex problems	**CP 6.4:** Ability to identify important and appropriate criteria and use themto evaluate available alternatives
CP 5.5: Effectively evaluate the success of solutions used in an attempt tosolve a complex problem	**CP 6.5:** Ability to select alternative(s) that adequately meet selected criteria
CP 5.6: Demonstrate a thorough understanding of a generalised conceptand facts specific to task or situation	**CP 6.6:** Preparedness to recognise and explain results after implementationof plans
CP 5.7: Provide new insight into controversial situation or task	**CP 6.7:** Implement strategies with accuracy
CP 5.8: Identify and prove misconceptions about a generalised concept orfact specific to a task or situation	
CP 5.9: Identify and explain a confusion, uncertainty or a contradiction surrounding an event	
CP 5.10: Develop and defend a logical plausible resolution to a confusion, uncertainty or contradiction surrounding an event	

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3. PERSONAL DEVELOPMENT AND LEADERSHIP (PL)

B7-B9	
PL5: PERSONAL DEVELOPMENT	**PL6: LEADERSHIP**
PL5.1: Understanding oneself (strengths, weaknesses, goals and aspirations)in reacting and adjusting to novel situations	**PL6.1:** Ability to serve group members effectively
PL5.2: Demonstrate a sense of belonging in a group	**PL6.2:** Division of tasks into solvable units and assigning group members totask units
PL5.3: Recognise one's emotional state and their preparedness to applyemotional intelligence	**PL6.3:** Ability to manage time effectively
PL5.4: Ability to understand one's personality traits	**PL6.4:** Ability to manage and resolve conflicts
PL5.5: Desire to accept one's true self and overcome weaknesses	**PL6.5:** Ability to monitor team members to ascertain progress
PL5.6: Ability to set and maintain personal standards and values	**PL6.6:** Ability to mentor peers
	PL6.7: Actively promote effective group interaction and the expression ofideas and opinions in a way that is sensitive to the feelings and background ofothers
	PL6.8: Actively assist group identify changes or modifications necessary inthe group activities and work towards carrying out those changes

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4. CULTURAL IDENTITY AND GLOBAL CITIZENSHIP (CG)

B7-B9	
CG5: CULTURAL IDENTITY	**CG6: GLOBAL CITIZENSHIP**
CG5.1: Show a strong sense of belonging to one's culture	**CG6.1:** Understanding of influences of globalisation on traditions, languagesand cultures
CG5.2: Develop and exhibit ability to defend one's cultural beliefs, practicesand norms	**CG6.2:** Recognise resistance to global practices that are inimical to ourculture
CG5.3: Develop and express respect, recognition and appreciation ofothers' cultures	**CG6.3:** Know the global discourse about the roles of males and females
CG5.4: Develop and exhibit a sense of cultural identity	**CG6.4:** Exhibit a sense of nationality and global identity
CG5.5: Adjust to the demands of customs, traditions, values and attitudes of society	

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5. CREATIVITY AND INNOVATION (CI)

B7-B9	
CI5: KNOWLEDGE, UNDERSTANDING, SKILLS AND STRATEGIES	**CI6: REFLECTION AND EVALUATION**
CI 5.1: Examine alternatives in creating new things	**CI 6.1:** Exhibit strong memory, intuitive thinking and respond appropriately
CI 5.2: Ability to merge simple/complex ideas to create novel situations orthings	**CI 6.2:** Ability to reflect on approaches to creative tasks and evaluate the effectiveness of tools used
CI 5.3: Identification of requirements of a given situation and justification ofmore than one creative tool that will be suitable	**CI 6.3:** Ability to select the most effective creative tools for work, and give reasons for the choice
CI 5.4: Ability to visualise alternatives, see possibilities and identify problemsand challenges	**CI 6.4:** Imagining and seeing things in a different way
CI 5.5: Ability to try new alternatives and different approaches	**CI 6.5:** Anticipate and overcome difficulties relating to taking initiatives
CI 5.6: Understand and use analogies and metaphors	**CI 6.6:** Being open-minded, adapting and modifying ideas to achieve creativeresults
CI 5.7: Putting forward constructive comments, ideas, explanations and newways of doing things	**CI 6.7:** Look and think about things differently and from different perspectives
	CI 6.8: Recognise and generalise information and experience; search fortrends and patterns
	CI 6.9: Interpret and apply learning in new contexts
	CI 6.10: Reflect on work and explore the thinking behind thoughts and processes

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6. DIGITAL LITERACY (DL)

B7-B9	
DL5: PHOTO-VISUAL AND INFORMATION LITERACY	**DL6: SOCIO-EMOTIONAL AND REPRODUCTION LITERACY**
DL5.1: Ability to ascertain when information is needed and be able toidentify, locate, evaluate and effectively use it to solve a problem	**DL 6.1:** Understand the sociological and emotional aspects of cyberspace
DL5.2: Ability to recognise and avoid traps in cyberspace	**DL 6.2:** Create a meaningful and original piece of work, or its interpretationby integrating existing information
DL5.3: Ability to find and utilise digital content	**DL6.3:** Use digital tools to create novel things
DL5.4: Ability to construct knowledge from a non-linear hyper-textualnavigation	**DL6.4:** Adhere to behavioural protocols that prevail in cyberspace
DL5.5: Evaluate the quality and validity of information	**DL6.5:** Recognition of societal issues emanating from the use of digitaltechnologies
DL5.6: Preparedness to make better decisions using available information	**DL6.6:** Knowledge and recognition of ethical use of information

Please note these inclusivity issues

The core competencies outlined in this document must be assessed taking into consideration learnerswith special needs (physical disabilities, learning disabilities, etc.).
Consider the use of realia for visual and visually challenged

learners.A system of creating alternatives for tasks must also be

adopted.

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